COMP1870 Theoretical Foundations of Computer Science I

School of Computer Science, University of Leeds

Worksheet - Week 1.4 submission template

Questions

7. Let A be the set $\{1, 2, 3, 4, 5, 6\}$ and let f and g be the following functions from A to A.

$$f = \{(1, 4), (2, 2), (3, 6), (4, 5), (5, 1), (6, 3)\} \qquad g = \{(1, 3), (2, 1), (3, 5), (4, 2), (5, 4), (6, 5)\}$$

   (a) Write down the ordered pairs in the inverse function f^{-1}.

   (b) Use your answer to (a) and function composition to find a function $h : A \to A$ such that $f \circ h = g$.

Write your solution in this box:

11. Let A be the set $\{1, 2, 3, 4, 5\}$. Let $R \subseteq A \times A$ be the following relation on A.

$$\{(1, 1), (1, 3), (2, 2), (2, 4), (3, 3), (3, 5), (4, 4), (5, 5)\}.$$

(a) Draw the arrow diagram of R.

(b) Is R reflexive? Justify your answer.

(c) Is R symmetric? Justify your answer.

(d) Is R anti-symmetric? Justify your answer.

(e) Is R transitive? Justify your answer.

Write your solution in this box: